

ITA1412 -ETHICAL HACKING

Experiment no:1

Experiment Name: Port Scanning Tools

Procedure:

Step 1: Open Nmap from Kali Linux (Go to Applications->select Information Gathering->select

Nmap)

Step 2: Perform different types of scans9`

(Tcp, Udp, Ack, Syn, Fin, Null, Xmas, Rpc, Idle)- scan types

Scanning Techniques

Flag Use Example

-sS TCP syn port scan nmap -sS 192.168.1.1

-sT TCP connect port scan nmap -sT 192.168.1.1

–sU UDP port scan nmap –sU 192.168.1.1

–sA TCP ack port scan nmap –sA 192.168.1.1

Output:

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ ping google.com  
PING google.com (142.251.220.110) 56(84) bytes of data.  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=64 time=39.3 ms  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=64 time=45.8 ms  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=64 time=76.2 ms  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=64 time=97.1 ms  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=5 ttl=64 time=108 ms  
^C  
--- google.com ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4023ms  
rtt min/avg/max/mdev = 39.329/73.263/107.811/27.110 ms  
(kali@kali)-[~]  
$ sudo apt update  
[sudo] password for kali:  
Hit:1 http://http.kali.org/kali kali-rolling InRelease  
731 packages can be upgraded. Run 'apt list --upgradable' to see them.  
(kali@kali)-[~]  
$ sudo apt install nmap  
nmap is already the newest version (7.99+dfsg-1kali1).  
nmap set to manually installed.  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 731  
(kali@kali)-[~]  
$ nmap --version  
Nmap version 7.99 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6 nmap-libdnet  
-1.18.0 ipv6  
Compiled without:  
Available nsock engines: epoll poll select  
(kali@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 85270sec preferred_lft 85270sec  
    inet6 fd17:625c:f037:2:18ac:924a:5755:8a7b/64 scope global dynamic noprefixroute  
        valid_lft 86253sec preferred_lft 14253sec  
    inet6 fe80::6151:3a26:a3f:762e/64 scope link noprefixroute
```

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: command not found: history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 85270sec preferred_lft 85270sec  
    inet6 fd17:625c:f037:2:18ac:924a:5755:8a7b/64 scope global dynamic noprefixroute  
        valid_lft 86253sec preferred_lft 14253sec  
    inet6 fe80::6151:3a26:a3f:762e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kali@kali)-[~]  
$ nmap 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 00:00 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000040s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.22 seconds  
(kali@kali)-[~]  
$ nmap -sV 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 00:01 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000033s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.54 seconds  
(kali@kali)-[~]  
$ nmap -p- 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 00:01 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000020s latency).  
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 65535 closed tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds  
(kali@kali)-[~]  
$
```

Experiment no:2

Experiment Name: Host Discovery

Procedure:

Step 1: Open Nmap from Kali Linux (Go to Applications->select Information Gathering->select Nmap)

Step 2: Perform different types of scans

(Tcp, Udp, Ack, Syn, Fin, Null, Xmas, Rpc, Idle)- scan types

To perform host discovery

-Pn only port scan nmap -Pn192.168.1.1

-sn only host discover nmap -sn192.168.1.1

-PR arp discovery on a local network nmap -PR192.168.1.1

-n disable DNS resolution nmap -n 192.168.1.1

Output:

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~ x kali@kali: ~ x  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)~  
$ nmap --version  
Nmap version 7.99 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6 nmap-libdnet-  
1.18.0 ipv6  
Compiled without:  
Available nsock engines: epoll poll select  
(kali@kali)~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
inet 127.0.0.1/8 scope host lo  
valid_lft forever preferred_lft forever  
inet6 ::1/128 scope host noprefixroute  
valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
valid_lft 84878sec preferred_lft 84878sec  
inet6 fd17:625c:f037:2:18ac:924a:5755:8a7b/64 scope global dynamic noprefixroute  
valid_lft 86291sec preferred_lft 14291sec  
inet6 fe80::6151:3a26:a3f:762e/64 scope link noprefixroute  
valid_lft forever preferred_lft forever  
(kali@kali)~  
$ nmap -sn 10.0.2.15/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 00:10 -0400  
Nmap scan report for 10.0.2.2  
Host is up (0.00038s latency).  
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)  
Nmap scan report for 10.0.2.3  
Host is up (0.0012s latency).  
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)  
Nmap scan report for 10.0.2.15  
Host is up.  
Nmap done: 256 IP addresses (3 hosts up) scanned in 7.91 seconds  
(kali@kali)~  
$ ping 10.0.2.2  
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.  
64 bytes from 10.0.2.2: icmp_seq=1 ttl=64 time=1.32 ms  
64 bytes from 10.0.2.2: icmp_seq=2 ttl=64 time=1.28 ms  
64 bytes from 10.0.2.2: icmp_seq=3 ttl=64 time=0.777 ms  
64 bytes from 10.0.2.2: icmp_seq=4 ttl=64 time=2.34 ms  
64 bytes from 10.0.2.2: icmp_seq=5 ttl=64 time=1.19 ms
```

```
(kali㉿kali)-[~]  
$ ping 10.0.2.2  
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.  
64 bytes from 10.0.2.2: icmp_seq=1 ttl=64 time=1.32 ms  
64 bytes from 10.0.2.2: icmp_seq=2 ttl=64 time=1.28 ms  
64 bytes from 10.0.2.2: icmp_seq=3 ttl=64 time=0.777 ms  
64 bytes from 10.0.2.2: icmp_seq=4 ttl=64 time=2.34 ms  
64 bytes from 10.0.2.2: icmp_seq=5 ttl=64 time=1.19 ms  
^C  
— 10.0.2.2 ping statistics —  
5 packets transmitted, 5 received, 0% packet loss, time 4026ms  
rtt min/avg/max/mdev = 0.777/1.379/2.337/0.516 ms
```

```
(kali㉿kali)-[~]  
$ hostname  
kali
```

```
(kali㉿kali)-[~]  
$ netstat -rn  
Kernel IP routing table  
Destination      Gateway          Genmask          Flags      MSS Window  irtt Iface  
0.0.0.0          10.0.2.2        0.0.0.0          UG          0 0        0 eth0  
10.0.2.0         0.0.0.0         255.255.255.0    U           0 0        0 eth0
```

Experiment no:3

Experiment Name: Cracking The Password Using The Hydra

Procedure

Step 1: To open it, go to Applications → Password Attacks → Online Attacks.: hydra → In this case, we will brute force FTP service of Metasploit able machine, which has IP 192.168.1.101

We have created in Kali a word list with extension 'lst' in the path `usr\share\wordlist\Metasploit`. The command will be as follows –

`hydra -l /usr/share/wordlists/metasploit/user -P`

`/usr/share/wordlists/metasploit/ passwords ftp://192.168.1.101 -V`

where –V is the username and password while trying

the username and password are found which are msfadmin: msfadmin

Output:

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~ kali@kali: ~ kali@kali: ~  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)~  
$ hydra -h  
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organi  
zations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
Syntax: hydra [[-l LOGIN]-L FILE] [-p PASS]-P FILE]] | [-C FILE]] [-e nsr] [-o FILE] [-t TASKS] [-M FILE [-T T  
ASKS]] [-w TIME] [-W TIME] [-f] [-s PORT] [-x MIN:MAX:CHARSET] [-c TIME] [-ISOuvVd46] [-m MODULE_OPT] [service:  
//server[:PORT]][/OPT]]  
  
Options:  
-R restore a previous aborted/crashed session  
-I ignore an existing restore file (don't wait 10 seconds)  
-S perform an SSL connect  
-s PORT if the service is on a different default port, define it here  
-l LOGIN or -L FILE login with LOGIN name, or load several logins from FILE  
-p PASS or -P FILE try password PASS, or load several passwords from FILE  
-x MIN:MAX:CHARSET password brute-force generation, type "-x -h" to get help  
-y disable use of symbols in brute-force, see above  
-r use a non-random shuffling method for option -x  
-e nsr try "n" null password, "s" login as pass and/or "r" reversed login  
-u loop around users, not passwords (effective! implied with -x)  
-C FILE colon separated "login:pass" format, instead of -L/-P options  
-M FILE list of servers to attack, one entry per line, ':' to specify port  
-D XoFY Divide wordlist into Y segments and use the Xth segment.  
-o FILE write found login/password pairs to FILE instead of stdout  
-b FORMAT specify the format for the -o FILE: text(default), json, jsonv1  
-f / -F exit when a login/pass pair is found (-M: -f per host, -F global)  
-t TASKS run TASKS number of connects in parallel per target (default: 16)  
-T TASKS run TASKS connects in parallel overall (for -M, default: 64)  
-w / -W TIME wait time for a response (32) / between connects per thread (0)  
-c TIME wait time per login attempt over all threads (enforces -t 1)  
-4 / -6 use IPv4 (default) / IPv6 addresses (put always in [] also in -M)  
-v / -V / -d verbose mode / show login+pass for each attempt / debug mode  
-O use old SSL v2 and v3  
-K do not redo failed attempts (good for -M mass scanning)  
-q do not print messages about connection errors  
-U service module usage details  
-m OPT options specific for a module, see -U output for information  
-h more command line options (COMPLETE HELP)  
server the target: DNS, IP or 192.168.0.0/24 (this OR the -M option)  
service the service to crack (see below for supported protocols)  
OPT some service modules support additional input (-U for module help)  
  
Supported services: adam6500 asterisk cisco cisco-enable cobaltstrike cvs firebird ftp[s] http[s]-{head|get|pos  
t} http[s]-{get|post}-form http-proxy http-proxy-urllenum icq imap[s] irc ldap2[s] ldap3[-{cram|digest}md5][s] m  
emcached mongodb mssql mysql nntp oracle-listener oracle-sid pcanywhere pcnfs pop3[s] postgres radmin2 rdp redi  
s rexec rlogin rpcap rsh rtsp s7-300 sip smb smb2 smtp[s] smtp-enum snmp socks5 ssh sshkey svn teamspeak telnet  
[s] vmauthd vnc xmpp
```

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~ x kali@kali: ~ x kali@kali: ~ x  
t} http[s]-{get|post}-form http-proxy http-proxy-urlenum icq imap[s] irc ldap2[s] ldap3[-{cram|digest}md5][s] m  
emcached mongodb mssql mysql nntp oracle-listener oracle-sid pcanywhere pcnfs pop3[s] postgres radmin2 rdp redi  
s rexec rlogin rpcap rsh rtsp s7-300 sip smb smb2 smtp[s] smtp-enum snmp socks5 ssh sshkey svn teamspeak telnet  
[s] vmauthd vnc xmpp  
  
Hydra is a tool to guess/crack valid login/password pairs.  
Licensed under AGPL v3.0. The newest version is always available at;  
https://github.com/vanhauser-thc/thc-hydra  
Please don't use in military or secret service organizations, or for illegal  
purposes. (This is a wish and non-binding - most such people do not care about  
laws and ethics anyway - and tell themselves they are one of the good ones.)  
These services were not compiled in: afp ncp oracle sapr3.  
  
Use HYDRA_PROXY_HTTP or HYDRA_PROXY environment variables for a proxy setup.  
E.g. % export HYDRA_PROXY=socks5://l:p@127.0.0.1:9150 (or: socks4:// connect://)  
% export HYDRA_PROXY=connect_and_socks_proxylist.txt (up to 64 entries)  
% export HYDRA_PROXY_HTTP=http://login:pass@proxy:8080  
% export HYDRA_PROXY_HTTP=proxylist.txt (up to 64 entries)  
  
Examples:  
hydra -l user -P passlist.txt ftp://192.168.0.1  
hydra -L userlist.txt -p defaultpw imap://192.168.0.1/PLAIN  
hydra -C defaults.txt -6 pop3s://[2001:db8::1]:143/TLS:DIGEST-MD5  
hydra -l admin -p password ftp://[192.168.0.0/24]/  
hydra -L logins.txt -P pws.txt -M targets.txt ssh  
  
(kali@kali)-[~]  
$ hydra --version  
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organi  
zations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
hydra: invalid option -- '-'  
  
(kali@kali)-[~]  
$ nano users.txt  
  
(kali@kali)-[~]  
$ cat users.txt  
student
```

```
↑ nano users.txt  
  
(kali@kali)-[~]  
$ cat users.txt  
student  
  
(kali@kali)-[~]  
$ nano passwords.txt  
  
(kali@kali)-[~]  
$ cat passwords.txt  
123456  
password  
student  
admin  
kali  
  
(kali@kali)-[~]  
$ hydra -U ssh  
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organi  
zations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-20 00:18:49  
  
Help for module ssh:  
=====
```

The Module ssh does not need or support optional parameters

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~ kali@kali: ~ kali@kali: ~  
GNU nano 9.0 passwords.txt  
123456  
password  
student  
admin  
kali  
  
[ Read 5 lines ]  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_/ Go To Line M-E Redo
```


Experiment no:4

Experiment Name: Information Gathering Using The Harvester

Procedure:

STEP 1: Open Terminal in the kali Linux

STEP 2: Run the command

Output:

[illegible]

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~ x kali@kali: ~ x kali@kali: ~ x kali@kali: ~ x  
-n, --dns-lookup Enable DNS server lookup, default False.  
-c, --dns-brute Perform a DNS brute force on the domain.  
-f, --filename FILENAME Save the results to an XML and JSON file.  
-w, --wordlist WORDLIST Specify a wordlist for API endpoint scanning.  
-a, --api-scan Scan for API endpoints.  
-q, --quiet Suppress missing API key warnings and reading the api-keys file.  
-b, --source SOURCE baidu, bevigil, bitbucket, brave, bufferoverrun, builtwith, censys, certspotter, chaos,  
commoncrawl, criminalip, crtsh, dehashed, dsndumpster, duckduckgo, fofa, fullhunt, github-  
code, gitlab, hackertarget, haveibeenpwned, hudsonrock, hunter, hunterhow, intelx, leakix,  
leaklookup, netlas, onyphe, otx, pentesttools, projectdiscovery, rapiddns, robtex,  
rocketreach, securityscorecard, securityTrails, shodan, subdomaincenter,  
subdomainfinderc99, thc, threatcrowd, tomba, urlscan, venacius, virustotal, waybackarchive,  
whoisxml, windvane, yahoo, zoomeye  
  
(kali@kali)-[~]  
$ theHarvester --version  
Read proxies.yaml from /etc/theHarvester/proxies.yaml  
*****  
* theHarvester 4.10.1 *  
* Coded by Christian Martorella *  
* Edge-Security Research *  
* cmartorella@edge-security.com *  
* *  
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-S START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t]  
[-r [DNS_RESOLVE]] [-n] [-c] [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]  
theHarvester: error: the following arguments are required: -d/--domain  
  
(kali@kali)-[~]  
$ theHarvester -h  
Read proxies.yaml from /etc/theHarvester/proxies.yaml  
*****  
* theHarvester 4.10.1 *  
* Coded by Christian Martorella *  
* Edge-Security Research *
```


Experiment no :5

Experiment Name: Use Google & Whois For Reconnaissance

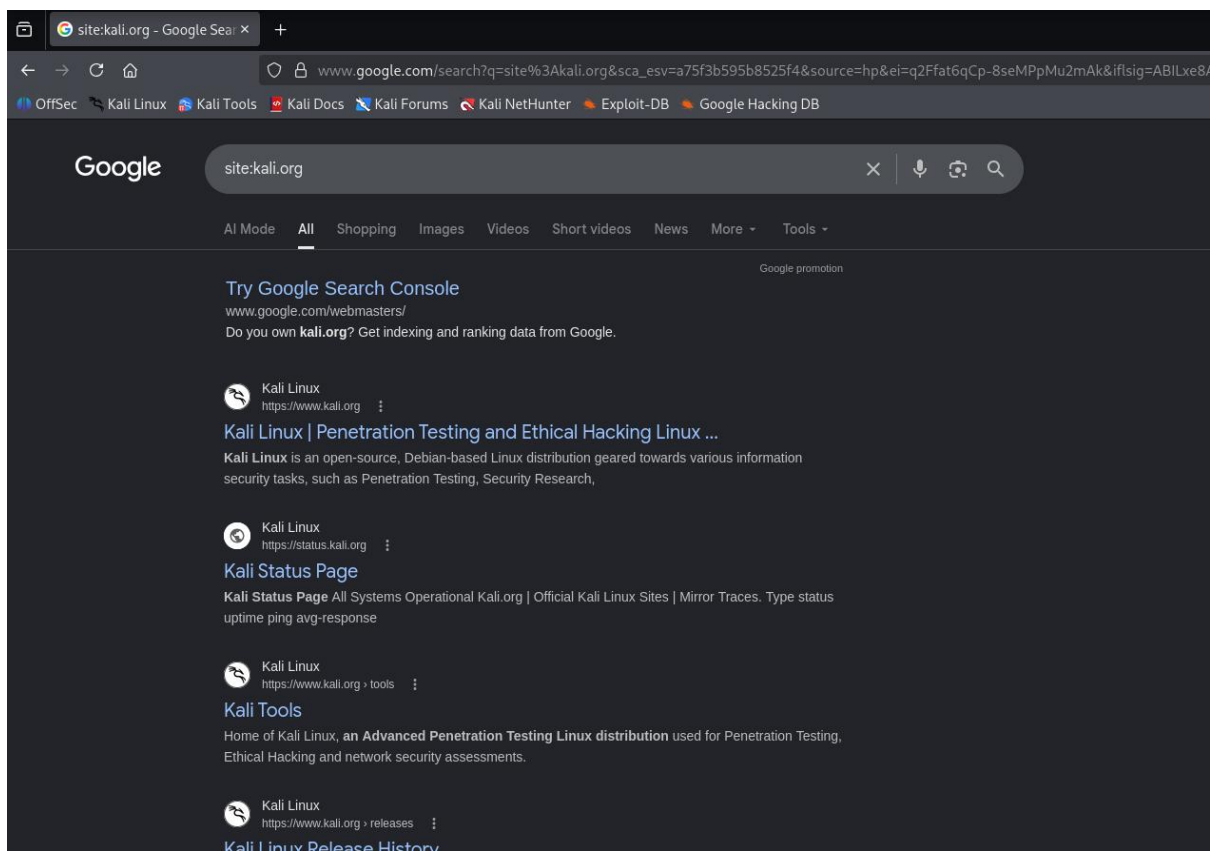
Procedure:

Step 1: In windows operating system opening google chrome & searching for whois website

Step 2: In whois website entering the www.saveetha.com

Step 3: Finally, we get the information of the website

Output:





site:kali.org filetype:pdf



ALL



SEARCH

IMAGES

VIDEOS

MAPS

NEWS

MORE

About 109,000 results



ilovepdf.com
<https://www.ilovepdf.com>

PDF to WORD | Convert PDF to Word online for free - iLovePDF

Convert PDF to editable Word documents for free. PDF to Word conversion is fast, secure and almost 100% accurate. Convert scanned PDF to DOC keeping the layout.



smallpdf.com
<https://smallpdf.com>

PDF to Word Converter | Convert PDF to Word Online for Free

Convert PDF to Word online for free. Turn your PDFs into fully editable Word documents in seconds. No signup required, no watermarks.

Reviews: 1M



freeconvert.com
<https://www.freeconvert.com>

PDF to WORD Converter - FreeConvert.com

PDF to WORD converter. Best way to convert PDF to WORD online at the highest quality. This tool is free, secure, and works on any web browser.



adobe.com
<https://www.adobe.com/acrobat/online>

Convert PDF to Word for free - PDF to DOCX Converter - Adobe

Adobe Acrobat's free online PDF to Word tool lets you drag and drop a PDF above and download an editable Word file for free — no sign-up, credit card, or subscription required.



kali@kali: ~

Session Actions Edit View Help

zsh: corrupt history file /home/kali/.zsh_history

(kali@kali)~

\$ whois --version

Version 5.6.6.

Report bugs to <md+whois@linux.it>.

(kali@kali)~

\$ sudo apt update

sudo apt install whois

[sudo] password for kali:

Get:1 http://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]

Get:2 http://mirrors.esto.network/kali kali-rolling/contrib Sources [72.8 kB]

Get:3 http://mirrors.esto.network/kali kali-rolling/main Sources [18.1 MB]

Get:4 http://mirrors.esto.network/kali kali-rolling/non-free-firmware Sources [10.4 kB]

Get:5 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [21.4 MB]

Get:6 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]

Get:7 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Packages [105 kB]

Get:8 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Contents (deb) [190 kB]

Get:9 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Packages [15.8 kB]

Get:10 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Contents (deb) [40.5 kB]

Fetched 93.4 MB in 40s (2,311 kB/s)

749 packages can be upgraded. Run 'apt list --upgradable' to see them.

whois is already the newest version (5.6.6).

Summary:

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 749

(kali@kali)~

\$ whois kali.org

connect: Connection refused

Experiment no :6

Experiment Name: Windows Operating System Commands Execution

Traceroute, Ping, Ifconfig & NETSTAT

Procedure:

Step 1: open windows command prompt and Type tracert command and type
tracert www.saveetha.com -> "Enter"

Step 2: Type ping command and type IP Address press "Enter"

Step 3: Type ifconfig command

step 4: type netstat

Output:

```
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mural>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . . : fe80::d853:f51b:3466:c8a6%6
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 12:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : saveetha.net
    Link-local IPv6 Address . . . . . : fe80::a10a:6262:be36:98ae%4
    IPv4 Address. . . . . : 172.23.29.25
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 172.23.16.1

C:\Users\mural>ping google.com

Pinging google.com [142.250.206.78] with 32 bytes of data:
Reply from 142.250.206.78: bytes=32 time=51ms TTL=117
Reply from 142.250.206.78: bytes=32 time=193ms TTL=117
Reply from 142.250.206.78: bytes=32 time=63ms TTL=117
Reply from 142.250.206.78: bytes=32 time=41ms TTL=117

Ping statistics for 142.250.206.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

```
Ping statistics for 142.250.206.78:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 41ms, Maximum = 193ms, Average = 87ms
```

```
C:\Users\mural>tracert google.com
```

```
Tracing route to google.com [142.250.206.78]
over a maximum of 30 hops:
```

1	20 ms	9 ms	11 ms	172.23.16.1
2	57 ms	36 ms	53 ms	172.22.8.1
3	58 ms	131 ms	40 ms	14.139.187.225
4	*	*	*	Request timed out.
5	*	*	*	Request timed out.
6	*	*	*	Request timed out.
7	*	*	*	Request timed out.
8	28 ms	20 ms	72 ms	72.14.195.128
9	*	24 ms	44 ms	142.251.227.213
10	62 ms	12 ms	16 ms	142.250.228.187
11	107 ms	98 ms	13 ms	pnmaaa-ay-in-f14.1e100.net [142.250.206.78]

```
Trace complete.
```

```
C:\Users\mural>netstat
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49684	murali:49685	ESTABLISHED
TCP	127.0.0.1:49685	murali:49684	ESTABLISHED
TCP	127.0.0.1:49686	murali:49687	ESTABLISHED
TCP	127.0.0.1:49687	murali:49686	ESTABLISHED
TCP	127.0.0.1:49690	murali:49691	ESTABLISHED
TCP	127.0.0.1:49691	murali:49690	ESTABLISHED
TCP	127.0.0.1:53263	murali:53264	ESTABLISHED
TCP	127.0.0.1:53264	murali:53263	ESTABLISHED
TCP	127.0.0.1:54215	murali:55169	ESTABLISHED
TCP	127.0.0.1:54217	murali:55169	ESTABLISHED
TCP	127.0.0.1:55169	murali:54215	ESTABLISHED
TCP	127.0.0.1:55169	murali:54217	ESTABLISHED
TCP	127.0.0.1:55169	murali:61106	ESTABLISHED
TCP	127.0.0.1:61106	murali:55169	ESTABLISHED
TCP	127.0.0.1:62267	murali:62268	ESTABLISHED
TCP	127.0.0.1:62268	murali:62267	ESTABLISHED
TCP	127.0.0.1:65252	murali:65253	ESTABLISHED
TCP	127.0.0.1:65253	murali:65252	ESTABLISHED
TCP	172.23.29.25:49362	20.195.84.7:https	ESTABLISHED
TCP	172.23.29.25:52428	243:https	ESTABLISHED

TCP	172.23.29.25:49362	20.195.84.7:https	ESTABLISHED
TCP	172.23.29.25:52428	243:https	ESTABLISHED
TCP	172.23.29.25:52910	93:https	ESTABLISHED
TCP	172.23.29.25:53265	ec2-3-111-241-224:https	ESTABLISHED
TCP	172.23.29.25:54405	172.64.155.209:https	ESTABLISHED
TCP	172.23.29.25:54963	ns31571428:https	ESTABLISHED
TCP	172.23.29.25:56118	4.213.25.241:https	ESTABLISHED
TCP	172.23.29.25:56734	52.110.15.186:https	ESTABLISHED
TCP	172.23.29.25:59859	sl-in-f188:5228	ESTABLISHED
TCP	172.23.29.25:62030	4.213.25.242:https	ESTABLISHED
TCP	172.23.29.25:62044	20.195.65.193:https	ESTABLISHED
TCP	172.23.29.25:62931	172.64.148.235:https	ESTABLISHED
TCP	172.23.29.25:62938	a104-86-189-120:https	ESTABLISHED
TCP	172.23.29.25:62939	a104-86-189-120:https	ESTABLISHED
TCP	172.23.29.25:62940	172.217.118.4:https	ESTABLISHED
TCP	172.23.29.25:62950	142.251.150.119:https	TIME_WAIT

```
C:\Users\mural>netstat -a
```

```
C:\Users\mural>netstat -a
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	murali:0	LISTENING
TCP	0.0.0.0:445	murali:0	LISTENING
TCP	0.0.0.0:5040	murali:0	LISTENING
TCP	0.0.0.0:49664	murali:0	LISTENING
TCP	0.0.0.0:49665	murali:0	LISTENING
TCP	0.0.0.0:49666	murali:0	LISTENING
TCP	0.0.0.0:49667	murali:0	LISTENING
TCP	0.0.0.0:49670	murali:0	LISTENING
TCP	0.0.0.0:49692	murali:0	LISTENING
TCP	127.0.0.1:24830	murali:0	LISTENING
TCP	127.0.0.1:49684	murali:49685	ESTABLISHED
TCP	127.0.0.1:49685	murali:49684	ESTABLISHED
TCP	127.0.0.1:49686	murali:49687	ESTABLISHED
TCP	127.0.0.1:49687	murali:49686	ESTABLISHED
TCP	127.0.0.1:49690	murali:49691	ESTABLISHED
TCP	127.0.0.1:49691	murali:49690	ESTABLISHED
TCP	127.0.0.1:54215	murali:55169	ESTABLISHED
TCP	127.0.0.1:54217	murali:55169	ESTABLISHED
TCP	127.0.0.1:55168	murali:0	LISTENING
TCP	127.0.0.1:55169	murali:0	LISTENING
TCP	127.0.0.1:55169	murali:54215	ESTABLISHED
TCP	127.0.0.1:55169	murali:54217	ESTABLISHED
TCP	127.0.0.1:55169	murali:61106	ESTABLISHED
TCP	127.0.0.1:61106	murali:55169	ESTABLISHED
TCP	127.0.0.1:62267	murali:62268	ESTABLISHED
TCP	127.0.0.1:62268	murali:62267	ESTABLISHED
TCP	127.0.0.1:62959	murali:62960	ESTABLISHED
TCP	127.0.0.1:62960	murali:62959	ESTABLISHED
TCP	127.0.0.1:65252	murali:65253	ESTABLISHED
TCP	127.0.0.1:65253	murali:65252	ESTABLISHED
TCP	172.23.29.25:139	murali:0	LISTENING
TCP	172.23.29.25:49362	20.195.84.7:https	ESTABLISHED
TCP	172.23.29.25:52428	243:https	ESTABLISHED
TCP	172.23.29.25:52910	93:https	ESTABLISHED
TCP	172.23.29.25:54405	172.64.155.209:https	ESTABLISHED
TCP	172.23.29.25:54963	ns31571428:https	ESTABLISHED
TCP	172.23.29.25:56118	4.213.25.241:https	ESTABLISHED
TCP	172.23.29.25:56734	52.110.15.186:https	ESTABLISHED
TCP	172.23.29.25:59859	sl-in-f188:5228	ESTABLISHED
TCP	172.23.29.25:62030	4.213.25.242:https	ESTABLISHED
TCP	172.23.29.25:62044	20.195.65.193:https	ESTABLISHED
TCP	172.23.29.25:62931	172.64.148.235:https	ESTABLISHED
TCP	172.23.29.25:62938	a104-86-189-120:https	ESTABLISHED
TCP	172.23.29.25:62939	a104-86-189-120:https	ESTABLISHED
TCP	172.23.29.25:62940	172.217.118.4:https	ESTABLISHED
TCP	172.23.29.25:62950	142.251.150.119:https	TIME_WAIT
TCP	172.23.29.25:62958	20.42.65.89:https	ESTABLISHED
TCP	172.23.29.25:62961	ec2-3-111-241-224:https	ESTABLISHED
TCP	172.23.29.25:62962	150.171.22.11:https	ESTABLISHED
TCP	192.168.56.1:139	murali:0	LISTENING
TCP	[::]:135	murali:0	LISTENING
TCP	[::]:445	murali:0	LISTENING
TCP	[::]:49664	murali:0	LISTENING
TCP	[::]:49665	murali:0	LISTENING
TCP	[::]:49666	murali:0	LISTENING
TCP	[::]:49667	murali:0	LISTENING
TCP	[::]:49670	murali:0	LISTENING
TCP	[::]:49692	murali:0	LISTENING


```

TCP      [::]:49666      murali:0      LISTENING
TCP      [::]:49667      murali:0      LISTENING
TCP      [::]:49670      murali:0      LISTENING
TCP      [::]:49692      murali:0      LISTENING
TCP      [::1]:42050      murali:0      LISTENING
TCP      [::1]:49671      murali:0      LISTENING
UDP      0.0.0.0:123      *:
UDP      0.0.0.0:5050      *:
UDP      0.0.0.0:5353      *:
UDP      0.0.0.0:5353      *:
UDP      0.0.0.0:5353      *:
UDP      0.0.0.0:5353      *:
UDP      0.0.0.0:5353      *:
UDP      0.0.0.0:5353      *:
UDP      0.0.0.0:5353      *:
UDP      0.0.0.0:5353      *:
UDP      0.0.0.0:5353      *:
UDP      0.0.0.0:49664      142.251.156.119:443
UDP      0.0.0.0:52465      *:
UDP      0.0.0.0:55266      *:
UDP      0.0.0.0:58732      172.217.24.138:443
UDP      127.0.0.1:1900      *:
UDP      127.0.0.1:51187      127.0.0.1:51187
UDP      127.0.0.1:61525      127.0.0.1:61525
UDP      127.0.0.1:62413      *:
UDP      127.0.0.1:62443      127.0.0.1:62444
UDP      127.0.0.1:62444      127.0.0.1:62443
UDP      172.23.29.25:137      *:
UDP      172.23.29.25:138      *:
UDP      172.23.29.25:1900      *:
UDP      172.23.29.25:62412      *:
UDP      192.168.56.1:137      *:
UDP      192.168.56.1:138      *:
UDP      192.168.56.1:1900      *:
UDP      192.168.56.1:62411      *:
UDP      [::]:123      *:
UDP      [::]:5353      *:
UDP      [::]:5353      *:
UDP      [::]:5353      *:
UDP      [::]:5353      *:
UDP      [::]:5353      *:
UDP      [::]:52465      *:
UDP      [::]:55266      *:
UDP      [::1]:1900      *:
UDP      [::1]:62410      *:
UDP      [fe80::a10a:6262:be36:98ae%4]:1900      *:
UDP      [fe80::a10a:6262:be36:98ae%4]:62409      *:
UDP      [fe80::d853:f51b:3466:c8a6%6]:1900      *:
UDP      [fe80::d853:f51b:3466:c8a6%6]:62408      *:

```

C:\Users\mural>route print

Interface List

```

7...bc fc e7 48 be 62 .....Intel(R) Ethernet Connection (16) I219-V
6...0a 00 27 00 00 06 .....VirtualBox Host-Only Ethernet Adapter
21...5a 02 05 da 51 ea .....Microsoft Wi-Fi Direct Virtual Adapter
9...5e 02 05 da 51 ea .....Microsoft Wi-Fi Direct Virtual Adapter #3
4...58 02 05 da 51 ea .....Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
1.....Software Loopback Interface 1

```

IPv4 Route Table

Active Routes:

```
=====
IPv4 Route Table
=====
```

```
Active Routes:
```

Network	Destination	Netmask	Gateway	Interface	Metric
0.0.0.0		0.0.0.0	172.23.16.1	172.23.29.25	35
127.0.0.0		255.0.0.0	On-link	127.0.0.1	331
127.0.0.1		255.255.255.255	On-link	127.0.0.1	331
127.255.255.255		255.255.255.255	On-link	127.0.0.1	331
172.23.16.0		255.255.240.0	On-link	172.23.29.25	291
172.23.29.25		255.255.255.255	On-link	172.23.29.25	291
172.23.31.255		255.255.255.255	On-link	172.23.29.25	291
192.168.56.0		255.255.255.0	On-link	192.168.56.1	281
192.168.56.1		255.255.255.255	On-link	192.168.56.1	281
192.168.56.255		255.255.255.255	On-link	192.168.56.1	281
224.0.0.0		240.0.0.0	On-link	127.0.0.1	331
224.0.0.0		240.0.0.0	On-link	192.168.56.1	281
224.0.0.0		240.0.0.0	On-link	172.23.29.25	291
255.255.255.255		255.255.255.255	On-link	127.0.0.1	331
255.255.255.255		255.255.255.255	On-link	192.168.56.1	281
255.255.255.255		255.255.255.255	On-link	172.23.29.25	291

```
=====
Persistent Routes:
```

```
None
```

```
IPv6 Route Table
=====
```

```
Active Routes:
```

If	Metric	Network	Destination	Gateway
1	331	::1/128		On-link
6	281	fe80::/64		On-link
4	291	fe80::/64		On-link
4	291	fe80::a10a:6262:be36:98ae/128		On-link
6	281	fe80::d853:f51b:3466:c8a6/128		On-link
1	331	ff00::/8		On-link
6	281	ff00::/8		On-link
4	291	ff00::/8		On-link

```
=====
Persistent Routes:
```

```
None
```

1.....Software Loopback Interface 1

IPv4 Route Table

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	0.0.0.0	172.23.16.1	172.23.29.25	35
127.0.0.0	255.0.0.0	255.0.0.0	On-link	127.0.0.1	331
127.0.0.1	255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
172.23.16.0	255.255.240.0	255.255.240.0	On-link	172.23.29.25	291
172.23.29.25	255.255.255.255	255.255.255.255	On-link	172.23.29.25	291
172.23.31.255	255.255.255.255	255.255.255.255	On-link	172.23.29.25	291
192.168.56.0	255.255.255.0	255.255.255.0	On-link	192.168.56.1	281
192.168.56.1	255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	240.0.0.0	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	240.0.0.0	On-link	172.23.29.25	291
255.255.255.255	255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
255.255.255.255	255.255.255.255	255.255.255.255	On-link	172.23.29.25	291

Persistent Routes:

None

IPv6 Route Table

Active Routes:

If	Metric	Network	Destination	Gateway
1	331	::1/128		On-link
6	281	fe80::/64		On-link
4	291	fe80::/64		On-link
4	291	fe80::a10a:6262:be36:98ae/128		On-link
6	281	fe80::d853:f51b:3466:c8a6/128		On-link
1	331	ff00::/8		On-link
6	281	ff00::/8		On-link
4	291	ff00::/8		On-link

Persistent Routes:

None

Experiment no:7

Experiment Name: Vulnerabilities Analysis Using CGI Scanning With Nikto

Procedure:

Step 1: open a terminal window and type nikto -H and press enter

Step 2: Type nikto -h <website> Tuning x and press enter

Step 3: Nikto starts web server scanning with all tuning options enabled.

Step 4: In the terminal window type “nikto -h <website>-Cgidirs all”and hit enter

Step 5:Nikto will scan the webserver as it looks vulnerable CGI directories. It scans the webserverand list out the directories

Output:

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ ping google.com  
PING google.com (142.251.220.110) 56(84) bytes of data.  
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=64 time=455 ms  
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=64 time=276 ms  
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=64 time=45.1 ms  
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=64 time=209 ms  
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=5 ttl=64 time=89.0 ms  
^C  
--- google.com ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4034ms  
rtt min/avg/max/mdev = 45.063/214.602/454.648/145.538 ms  
(kali@kali)-[~]  
$ nikto -Version  
Nikto 2.6.0 (LW 2.5)  
(kali@kali)-[~]  
$ nikto -Help  
  
Options:  
-Add-header      Add HTTP headers (can be used multiple times, one per header pair)  
-ask+           Whether to ask about submitting updates  
                 yes   Ask about each (default)  
                 no    Don't ask, don't send  
                 auto  Don't ask, just send  
-check6         Check if IPv6 is working (connects to ipv6.google.com or value set in nikto.conf)  
-Cgidirs+       Scan these CGI dirs: "none", "all", or values like "/cgi/ /cgi-a/"  
-config+       Use this config file  
-Display+      Turn on/off display outputs:  
                 1     Show redirects  
                 2     Show cookies received  
                 3     Show all 200/OK responses  
                 4     Show URLs which require authentication  
                 D     Debug output  
                 E     Display all HTTP errors  
                 P     Print progress to STDOUT  
                 S     Scrub output of IPs and hostnames  
                 V     Verbose output  
-dbcheck        Check database and other key files for syntax errors  
-evasion+      Encoding technique:  
                 1     Random URI encoding (non-UTF8)  
                 2     Directory self-reference (../)  
                 3     Premature URL ending  
                 4     Prepend long random string  
                 5     Fake parameter  
                 6     TAB as request spacer  
                 7     Change the case of the URL  
                 8     Use Windows directory separator (\)  
                 A     Use a carriage return (0x0d) as a request spacer
```

```
kali@kali: ~  
Session Actions Edit View Help  
ge.html 7 Change the case of the URL  
8 Use Windows directory separator (\)  
A Use a carriage return (0x0d) as a request spacer  
B Use binary value 0x0b as a request spacer  
-followredirects Follow 3xx redirects to new location  
-Format+ Save file (-o) format:  
csv Comma-separated-value  
json JSON Format  
htm HTML Format  
sql Generic SQL (see docs for schema)  
txt Plain text  
xml XML Format  
(if not specified the format will be taken from the file extension passed to -o  
utput)  
-Help This help information  
-host+ Target host/URL  
-id+ Host authentication to use, format is id:pass or id:pass:realm  
-ipv4 IPv4 Only  
-ipv6 IPv6 Only  
-key+ Client certificate key file  
-list-plugins List all available plugins, perform no testing  
-maxtime+ Maximum testing time per host (e.g., 1h, 60m, 3600s)  
-mutate+ Guess additional file names:  
1 Test all files with all root directories  
2 Guess for password file names  
3 Enumerate user names via Apache (/~user type requests)  
4 Enumerate user names via cgiwrap (/cgi-bin/cgiwrap/~user type requests)  
6 Attempt to guess directory names from the supplied dictionary file  
-mutate-options Provide information for mutates  
-nocheck Don't check for updates on startup  
-nocookies Do not use cookies from responses in requests  
-nointeractive Disables interactive features  
-nolookup Disables DNS lookups  
-nossl Disables the use of SSL  
-noslash Strip trailing slash from URL (e.g., '/admin/' to '/admin')  
-no404 Disables nikto attempting to guess a 404 page  
-Option Over-ride an option in nikto.conf, can be issued multiple times  
-output+ Write output to this file ('.' for auto-name)  
-Pause+ Pause between tests (seconds)  
-Platform+ Platform of target (nix, win, all)  
-Plugins+ List of plugins to run (default: ALL)  
-port+ Port to use (default 80)  
-RSAcert+ Client certificate file  
-root+ Prepend root value to all requests, format is /directory  
-Save Save positive responses to this directory ('.' for auto-name)  
-ssl Force ssl mode on port  
-Tuning+ Scan tuning:  
1 Interesting File / Seen in logs  
2 Misconfiguration / Default File  
3 Information Disclosure  
4 Injection (XSS/Script/HTML)
```

```

ge.html
2 Misconfiguration / Default File
3 Information Disclosure
4 Injection (XSS/Script/HTML)
5 Remote File Retrieval - Inside Web Root
6 Denial of Service
7 Remote File Retrieval - Server Wide
8 Command Execution / Remote Shell
9 SQL Injection
0 File Upload
a Authentication Bypass
b Software Identification
c Remote Source Inclusion
d Webservice
e Administrative Console
x Reverse Tuning Options (i.e., include all except specified)

-timeout+ Timeout for requests (default 10 seconds)
-Userdbs Load only user databases, not the standard databases
        all Disable standard dbs and load only user dbs
        tests Disable only db_tests and load udb_tests
-useragent Force User-Agent instead of pulling from database
-url+ Target host/URL (alias of -host)
-useproxy Use the proxy defined in nikto.conf, or argument http://server:port
-Version Print plugin and database versions
-vhost+ Virtual host (for Host header)
-404code Ignore these HTTP codes as negative responses (always). Format is "302,301".
-404string Ignore this string in response body content as negative response (always). Can be a
regular expression.

```

+ requires a value

Want to know more about Kali? Search for it here!



```
(kali@kali)-[~]
```

```
$ nikto -h testphp.vulnweb.com
```

```
- Nikto v2.6.0
```

[Documentation](#)

[Kali Tools](#)

```
+ [FAIL] Unable to connect to testphp.vulnweb.com:80.
```

```
(kali@kali)-[~]
```

```
$ nikto -h https://testphp.vulnweb.com
```

```
- Nikto v2.6.0
```

```
+ [FAIL] Unable to connect to testphp.vulnweb.com:443.
```

```
(kali@kali)-[~]
```

```
$ apache2 -v
```

```
Server version: Apache/2.4.68 (Debian)
```

```
Server built: 2026-06-11T18:22:17
```

```
(kali@kali)-[~]
```

```
$ sudo systemctl start apache2
```

```
[sudo] password for kali:
```

Server built: 2026-06-11T18:22:17

```
(kali@kali)-[~]
$ sudo systemctl start apache2
[sudo] password for kali:
```

```
(kali@kali)-[~]
$ nikto -h http://127.0.0.1
- Nikto v2.6.0
```

```
+ Target IP: 127.0.0.1
+ Target Hostname: 127.0.0.1
+ Target Port: 80
+ Platform: Linux/Unix
+ Start Time: 2026-07-20 01:09:01 (GMT-4)

+ Server: Apache/2.4.68 (Debian)
+ ERROR: Failed to check for updates: 403
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
+ [999984] /: Server may leak inodes via ETags, header found with file /, inode: 29cf, size: 6545fc74d1156, mt
ime: gzip. See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/e
n-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en
-US/docs/Web/HTTP/Headers/X-Content-Type-Options
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org
/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/
docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/doc
s/Web/HTTP/Headers/Referrer-Policy
+ [999990] OPTIONS: Allowed HTTP Methods: GET, POST, OPTIONS, HEAD .
+ [001406] /server-status: The mod_status module reveals Apache information. See: https://httpd.apache.org/doc
s/2.4/mod/mod_status.html

+ [007342] /: X-Frame-Options header is deprecated and was replaced with the Content-Security-Policy HTTP head
er with the frame-ancestors directive. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Header
s/X-Frame-Options
+ [007352] /: The X-Content-Type-Options header is not set. This could allow the user agent to render the cont
ent of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-sca
nner/vulnerabilities/missing-content-type-header/
+ 8083 requests: 16 errors and 10 items reported on the remote host
+ End Time: 2026-07-20 01:15:03 (GMT-4) (362 seconds)
```

+ 1 host(s) tested

```
*****
Portions of the server's headers (Apache/2.4.68) are not in
the Nikto 2.6.0 database or are newer than the known string. Would you like
to submit this information (*no server specific data*) to CIRT.net
```

+ 1 host(s) tested

```
*****
Portions of the server's headers (Apache/2.4.68) are not in
the Nikto 2.6.0 database or are newer than the known string. Would you like
to submit this information (*no server specific data*) to CIRT.net
for a Nikto update (or you may email to sullo@cirt.net) (y/n)? y
```

- Sent updated info to cirt.net -- Thank you!